SEARCHING STRINGS

Alphabet: $\Sigma = \{\sigma_1, \sigma_2, \ldots, \sigma_r\}$ $\quad 0 < r < \infty$

$T : \Sigma[n]$ // text $\quad 0 < m \leq n < \infty$

$P : \Sigma[m]$ // pattern

$P[0..m) = P[0..m-1]$; $T[0..n) = T[0..n-1]$

s is a possible shift

D $s \in 0..n-m$ is valid shift of P on T **iff**

$$T[s..s+m) = P[0..m)$$

Problem: Determine the set of valid shifts:

$$V \stackrel{def}{=} \{ s \in \underbrace{0..n-m}_{\text{possible shifts}} \mid T[s..s+m) \stackrel{?}{=} P[0..m) \}$$

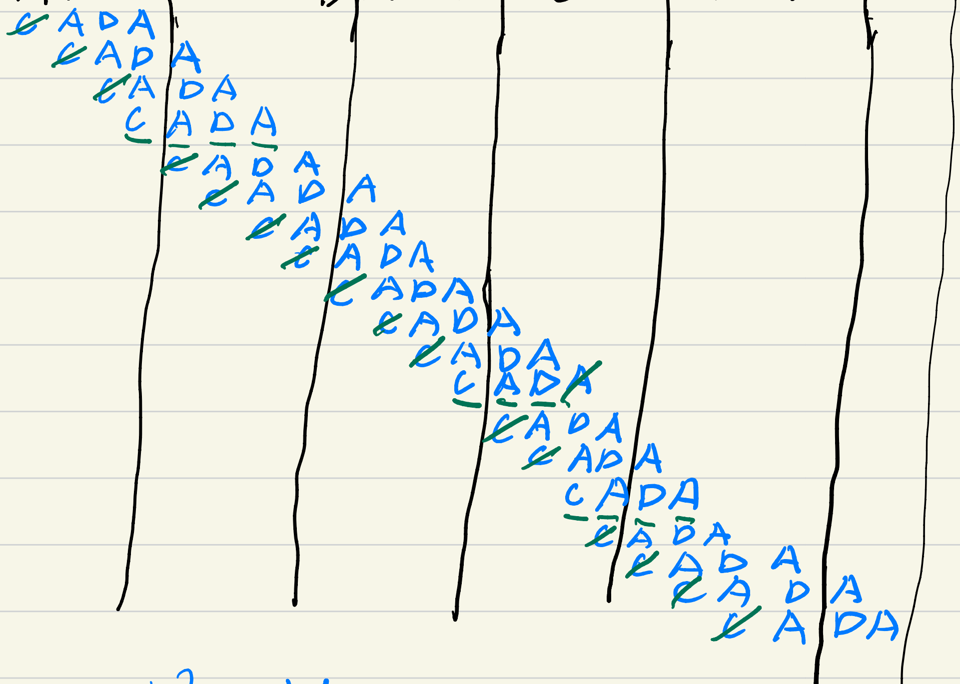
$P = CADA$

$T = ADACADABADACADCADABADA$ (positions marked 0, 4, 8, 12, 14, 16, 20)

Example

$n = 22$

$m = 4$



$s = 3$

$s = 14$

(Naive solution: we check each possible shift, if it is a valid shift.)

(19 possible shifts were checked)

$S = \{3, 14\} = V$

$\text{BruteForce}(T : \Sigma[n]\ ;\ P : \Sigma[m]\ ;\ S : \mathbb{N}\{\})$

- $S := \{\}$
- $s := 0$ **to** $n - m$
  - IF $\text{match}(T, s, P)$ // $T[s\,..\,s{+}m) = P[0\,..\,m)$
    - THEN: $S := S \cup \{s\}$
    - ELSE: SKIP

$\text{match}(\ T : \Sigma[]\ ;\ s : \mathbb{N}\ ;\ P : \Sigma[m]\ ) : \mathbb{B}$

- // check, whether $T[s\,..\,s{+}m) = P[0\,..\,m)$
- $j := 0$
- WHILE $j < m \wedge T[s + j] = P[j]$
  - $j{+}{+}$
- **return** $j \geq m$

Output: $S = V$

(NAIVE STRING MATCHING)

$mT_{match}(m) \in \Theta(1)$

$MT_{match}(m) \in \Theta(m)$

$mT_{BF}(n,m) \in \Theta(n - m + 1)$

$MT_{BF}(n,m) \in \Theta((n - m + 1) * m)$

$MT_{BF}(n) \in \Theta(n^2) \quad (m \approx \frac{n}{2})$

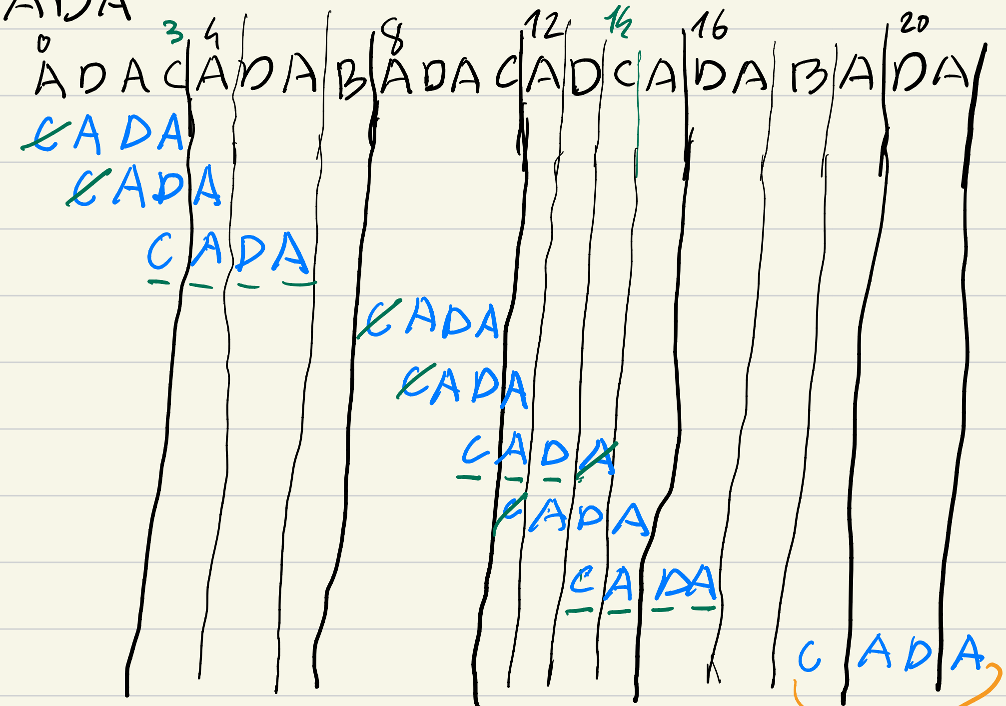
Quicksearch*

σ	A	B	C	D
shift(σ)	1	5	4	2

(see the next page)

$P = CADA$

$T = ADACADABADACADCADABADA$



$s = 3$

$s = 14$

This ($s=19$) is not a valid shift $\Rightarrow$ STOP

* A variant of Boyer-Moore

(8 possible shifts were checked.)

$\Sigma = \{A, B, C, D\}$ (for example) $(m = 4)$

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   Text: ...xxxxA......xxxxB......xxxxC......xxxxD...
Pattern:     CADA         CADA     CADA      CADA
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The appropriate *shift* values are given in the following table.

σ	A	B	C	D
$shift(\sigma)$	1	5	4	2

initShift$(P : \Sigma[m])$

$\forall \sigma \in \Sigma$

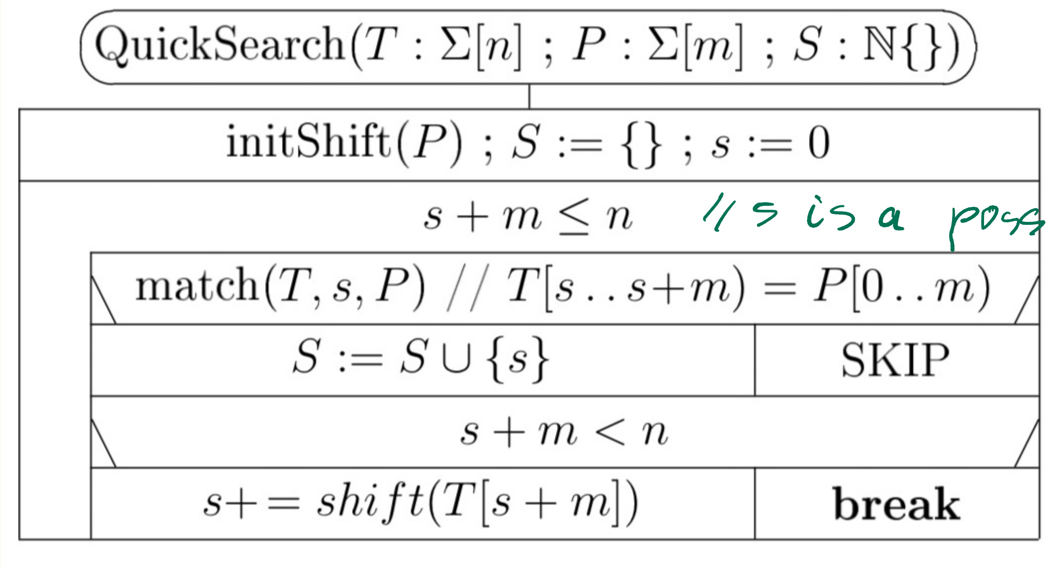
$\quad shift(\sigma) := m + 1$

$j := 0$ **to** $m-1$

$\quad shift(P[j]) := m - j$

$\sigma =$	A	B	C	D	
$shift(\sigma)=$	5	5	5	5	initial
C			4		
A	3				
D				2	
A	1				
$shift(\sigma)=$	1	5	4	2	final

$T(m) \in \Theta(m)$



If we want to optimize for the worst case, we need another algorithm

$$MT_{QS}(n,m) \in \Theta(m*(n-m+1)+m)$$

$$mT_{QS}(n,m) \in \Theta\left(\frac{n}{m+1}+m\right) \quad (\text{if } P \cap T = \emptyset)$$

Experience: $AT_{QS}(n,m)$ is much closer to the best case...

11-08 KNUTH-MORRIS-PRATT (KMP) algorithm

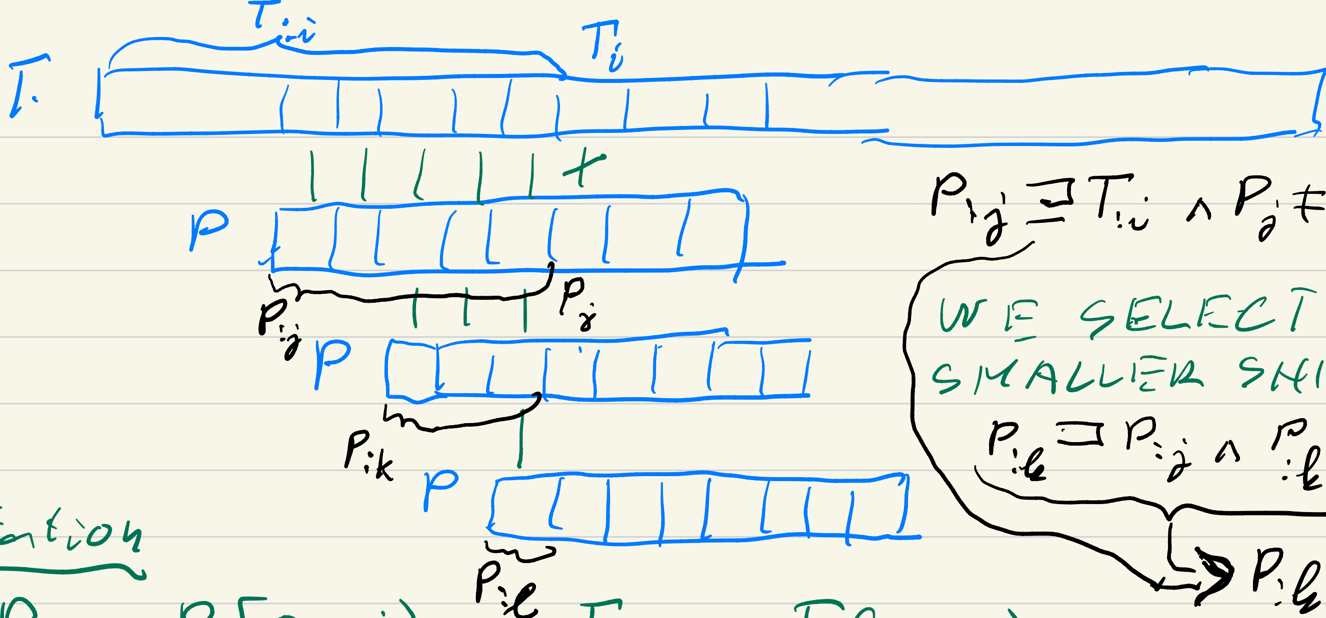
$mT(n,m), MT(n,m) \in \Theta(n)$

$\dots$	T_{i-5}	T_{i-4}	T_{i-3}	T_{i-2}	T_{i-1}	T_i	T_{i+1}	T_{i+2}
$\dots$	B	A	B	A	B	A	B	B
$P =$	$\underline{B}$	$\underline{A}$	$\underline{B}$	$\underline{A}$	$\underline{B}$	$\cancel{B}$		
			B	A	B	$\underline{A}$	$\underline{B}$	$\underline{B}$

$T : \Sigma[n]$ $\quad T[0..n)$

$p : \Sigma[m]$ $\quad P[0..m)$

$0 < m \leq n < \infty$



$P_{:j} \sqsupset T_{:i} \wedge P_j \neq T_i$

WE SELECT THE SMALLER SHIFT

$P_{:\ell} \sqsupset P_{:j} \wedge P_{:\ell} \sqsubset P_{:j}$!

$\Rightarrow P_{:\ell} \sqsupset T_{:i}$

$P_{:\ell} \sqsupset P_{:j} \wedge P_{:\ell} \sqsubset P_{:j}$

$\Downarrow$

$(\ell<k)$ $P_{:\ell} \sqsupset T_{:i}$

Notation

$P_{:j} = P[0..j)$ $\quad T_{:i} = T[0..i)$

$x \sqsubseteq y$ x is prefix of y

prefixes of BABA : ε, B, BA, BAB, BABA

$x \sqsubset y$: x is a proper prefix of y

Proper prefixes of BABA: ε, B, BA, BAB

$x \sqsupseteq y$: x is suffix of y

Suffixes of BABA: BABA, ABA, BA, A, ε

$x \sqsupset y$: x is proper suffix of y

Proper suffixes of BABA: ABA, BA, A, ε

$P_{:0} = \varepsilon$, $(BABA)_{:3} = BAB$

Problem: $P_{:j} \sqsupset T_{:i} \wedge P_j \neq T_i$: what to do

Solution: We need the longest proper prefix of $P_{:j}$ which is also suffix of $P_{:j}$

D. (prefix function $\pi : 1..m \to 0..m-1$)

$$\pi(j) = \max\{h \in 0..j-1 \mid P_{:h} \sqsupset P_{:j}\} \quad (j \in 1..m)$$

Th. $\pi(j) \in 0..j-1$

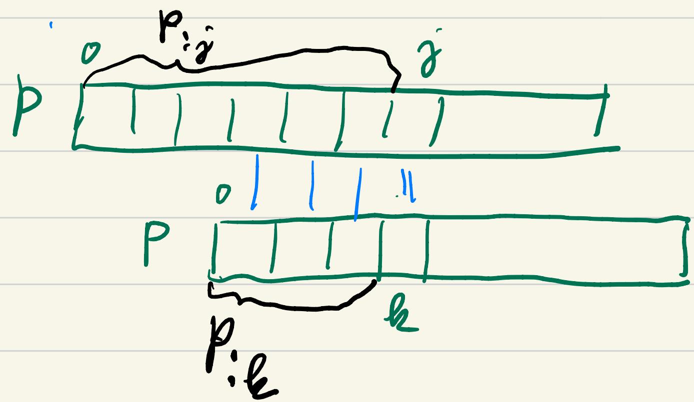
Th. $j \in 1..m-1 \Rightarrow \pi(j+1) \leq \pi(j)+1$

Proof *If* $\pi(j+1) = 0 \Rightarrow \pi(j+1) = 0 \leq 0+1 \leq \pi(j)+1$ ✓

If $\pi(j+1) > 0 \Rightarrow P_{:\pi(j+1)} = P_{:\pi(j+1)-1+1} \sqsupset P_{:j+1} \Rightarrow$

$\Rightarrow P_{:\pi(j+1)-1} \sqsupset P_{:j} \Rightarrow \pi(j+1)-1 \le \pi(j) \Rightarrow \dots$

<u>Lemma</u> $P_{:k+1} \sqsupset P_{:j+1} \iff P_{:k} \sqsupset P_{:j} \wedge P_k = P_j$



$\Rightarrow \underline{\underline{\pi(j+1) \le \pi(j)+1}}$

$P[j-1]=$	B	A	B	A	B	B	A	B
$j=$	1	2	3	4	5	6	7	8
$\pi(j)=$	0	0	1	2	3	1	2	3

0 3 4 8 12 16 20 24 28

T = B A B B A B A B B A B B A B A B B A A B A B A B B A B A B

B A B ~~A~~

B ~~A~~

s=3 B A B A B B A B

B A B ~~A~~

B ~~A~~

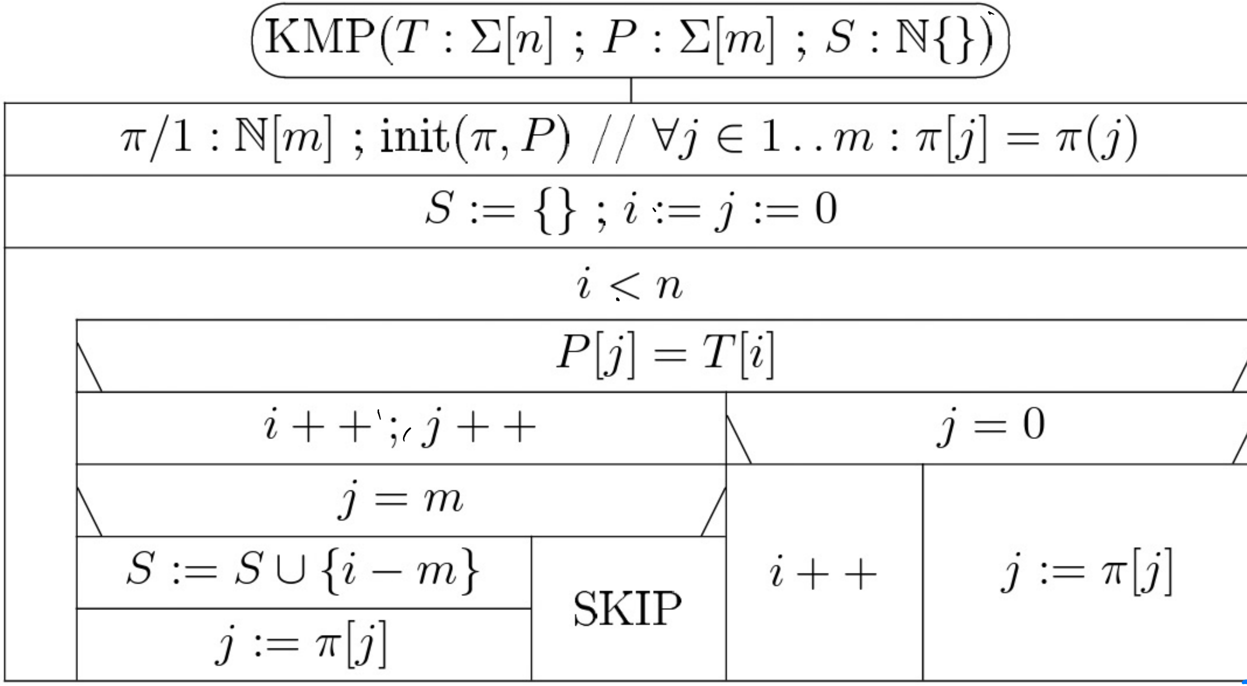
B A B A B B A ~~B~~

B A ~~B~~

~~B~~

s=19 B A B A B B A B

| BAB | A | B |



$\Theta(m)$ ← We prove this later

$0 \le i \le n$

$0 \le j \le m$

$j \le i$

$mT(n), MT(n) \in \Theta(n)$

$mT_{KMP}(n,m), MT_{KMP}(n,m) \in \Theta(n+m)$

$m \le n$

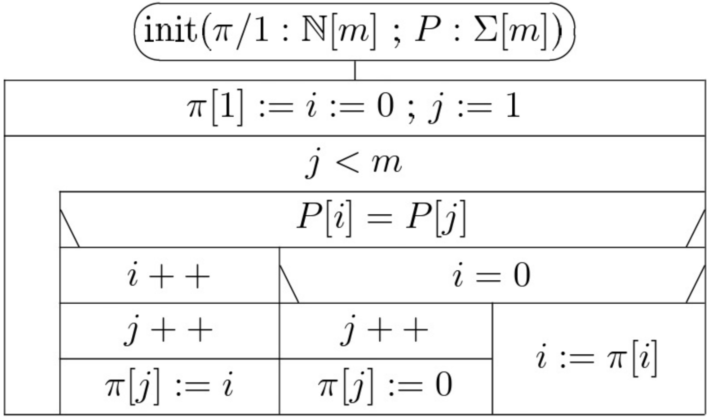
$k :=$ number of iterations $\Rightarrow k \ge n$

Let us see $2i - j \in 0..2n$

Before the 1st it.: $2i - j = 0$ $\Rightarrow k \le 2n$

$2i - j$ increases by $\forall$ it.

$\boxed{mT_{KMP}(n), MT_{KMP}(n) \in \Theta(n)}$



Invariant of the loop:
$P_{:i} \sqsupset P_{:j} \wedge$
$[\forall k \in 1..j : \pi[k] = \pi(k)\,]$
$\wedge\, 0 \le i < j \le m \,\wedge$
$(j < m \rightarrow \pi(j+1) \le i+1)$

$k \geq m-1$ $\begin{cases} \text{before 1st it.: } j = 1 \\ \forall \text{ it. increases } j \text{ by } 0 \text{ } \underline{\text{or}} \text{ } 1. \\ \text{at the end } j = m \end{cases}$

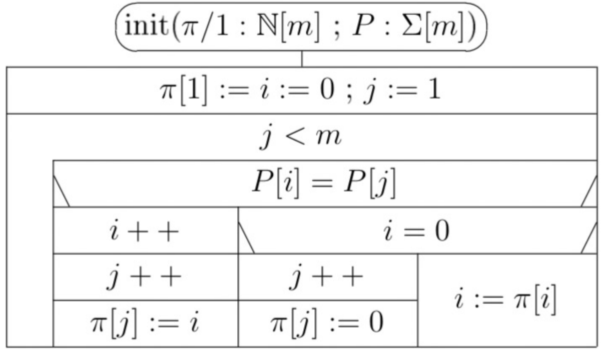
for max?

Let us see: $2j - i \in 2..2m$

$2j - i = j + (j - i)$

$\left.\begin{array}{l} \text{before 1st it.: } 2j - i = 2 \\ \forall \text{ it. increases } 2j - i \\ 2j - i \le 2m \end{array}\right\}$ $\left.\begin{array}{c} k \le 2m-2 \\ \vee\!\mid \\ m-1 \end{array}\right\}$ $\left\{\begin{array}{l} mT_{init}(n) \\ MT_{init}(n) \end{array}\right\} \Theta(m)$

The illustration of the $\text{init}(\pi, P)$ procedure on the pattern $ABABBABA$:
(We start a new line at the beginning of each of the three program branches.)



i	j	$\pi[j]$	0 A	1 B	2 A	3 B	4 B	5 A	6 B	7 A
0	1	0		$\cancel{A}$						
0	2	0			$\underline{A}$					
1	3	1			A	$\underline{B}$				
2	4	2			A	B	$\cancel{A}$			
0	4	2					$\cancel{A}$			
0	5	0						$\underline{A}$		
1	6	1						A	$\underline{B}$	
2	7	2						A	B	$\underline{A}$
3	8	3								

The result:

$P[j-1] =$	A	B	A	B	B	A	B	A
$j =$	1	2	3	4	5	6	7	8
$\pi[j] =$	0	0	1	2	0	1	2	3